

STOCK RETURN PREDICTION: Using Ridge and Lasso Regression with Feature Scaling

PROBLEM STATEMENT:

Stock market prediction is challenging because stock prices change continuously and are affected by many factors. In the RELIANCE stock dataset, features such as Open, High, Low, Close, and Adjusted Close are highly correlated. This can create multicollinearity, making traditional linear regression models unstable. The presence of large changes in daily returns can also make accurate prediction difficult.

To address these problems, this project uses feature scaling, Ridge Regression, and Lasso Regression for stock return prediction. Ridge and Lasso add regularization, which helps control large model coefficients and reduce the risk of overfitting. The project compares both models using MSE, MAE, and R^2 to determine which approach is more suitable for predicting RELIANCE stock returns.

DATASET IDENTIFICATION:

The project uses the RELIANCE.NS historical stock dataset, containing daily stock-market information for Reliance Industries. The dataset includes historical opening, high, low, closing, adjusted closing prices, and trading volume. It can be used to analyze stock-price movements and build a stock-return prediction model.

File	Description	Key Columns
RELIANCE.NS_1973-05-08_2025-03-01.csv	Daily historical stock-market data of Reliance Industries	date, open, high, low, close, adj close, volume
reliance_returns.csv	Dataset after calculating daily stock returns from historical prices	date, close, adj close, return
reliance_scaled.csv	Preprocessed dataset after applying feature scaling for machine-learning models	open, high, low, close, volume, return

The project uses the RELIANCE.NS historical stock dataset, which contains daily stock-market information for Reliance Industries. The dataset includes 7,324 records

and 7 columns: Date, Open, High, Low, Close, Adjusted Close, and Volume. It provides information about daily stock prices and trading activity and can be used to analyze stock-price movements and calculate daily returns for Ridge and Lasso Regression-based stock return prediction.

SUITABILITY OF THE DATASET:

The RELIANCE.NS historical stock dataset is suitable for this project because it contains daily stock-market features such as OPEN, HIGH, LOW, CLOSE, ADJUSTED CLOSE, AND VOLUME. These features provide useful information about stock-price movements and trading activity. The historical data can be used to calculate DAILY STOCK RETURNS, which can then be predicted using RIDGE AND LASSO REGRESSION. The dataset also contains numerical features that can benefit from FEATURE SCALING AND REGULARIZATION, making it appropriate for the proposed machine-learning approach

KNOWN LIMITATIONS:

- Limited features: The dataset mainly contains price and volume information.
- No direct return column: Daily stock return needs to be calculated from the price data.

- Single stock: The dataset focuses only on Reliance Industries.
- Possible class imbalance and home-ground data analysis (EDA).
- Correlated features: Open, high, low, and close prices are closely related.
- Lewis–Stern (DLS) method needs separate analysis.